YINAN FENG

PhD Candidate in Geography | Remote Sensing & GeoAI

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Geospatial researcher developing open building datasets and interactive tools with aerial imagery, elevation, machine learning, and large-scale spatial processing.

SELECTED PROJECTS

US Building Footprint (USBF) | Data & research

Developed selective refinement of GlobalBuildingAtlas footprints using aerial imagery and elevation. Published a six-state update with 25.1 million buildings, retaining source heights, original geometry, identifiers, and edit provenance.

Dataset Reproducible workflow

US Buildings | Geospatial website

Designed and built a browser GIS for footprint and height exploration, dataset comparison, available-height 3D display, and data export. Connected live USBF data and added local imports, scoped analysis, and map export.

Open website

RESEARCH EXPERIENCE

Graduate Research Intern | Oak Ridge National Laboratory 2025

Worked with the NBEM team on building-footprint integration and calibration for AutoBEM and the downstream effects of footprint geometry on building energy modeling.

Research Intern | Chinese Academy of Sciences 2021 - 2022

Analyzed GRACE and MODIS observations of water resources, grain-planting patterns, and crop-residue burning; contributed data processing, visualization, and scientific writing.

EDUCATION

University of Tennessee, Knoxville 2023 - Present

PhD in Geography (in progress) | Advisor: Qiusheng Wu

The Ohio State University 2018 - 2020

MS, Environmental Science | Advisor: Kaiguang Zhao

Liaoning University of Technology 2014 - 2018

BS, Environmental Science

SELECTED PUBLICATIONS

Feng, Y. & Wang, J. (2022). Detecting the spatial mismatch of water resources and grain planting pattern changes in China based on satellite data. Frontiers in Environmental Science.

Hu, T., et al., including Feng, Y. (2021). Mapping fine-scale human disturbances in a working landscape with Landsat time series on Google Earth Engine. ISPRS Journal of Photogrammetry and Remote Sensing.

SKILLS

Python, R, GIS, remote sensing, Google Earth Engine, deep learning, geospatial visualization, high-performance computing, reproducible data workflows.